Homework 2

Due date: Sunday July 5th at 11:59 PM ET

You can submit your homework on LEARN in PDF format only (1 doc.) and submit your code as a .ipynb file

Problem 1

Consider a star with a radius R_{star} with a planet on a circular orbit of semi-major axis a. Suppose that the inclination of the system is randomly distributed with respect to an observer on Earth, and that the planet has a radius much smaller than that of the star, and show that the probability P that the observer can see planetary transits from that planet is given by:

$$P \propto \frac{R_{star}}{a}$$

Problem 2

a) Use a programming language of your choice to write a code that automatically downloads the latest data from the NASA exoplanet archive to plot a histogram of the total number of confirmed exoplanets discovered every year. This code should allow you to update that figure at any time in the future by just launching it. You can find help on accessing the NASA exoplanet archive programmatically with simple URL queries at https://exoplanetarchive.ipac.caltech.edu/docs/program_interfaces.html

b) Why are there two large peaks in recent years where a lot of exoplanets were discovered ?
Provide your code in an annex, as well as your figure.

Problem 3

Take a real TESS light curve of a confirmed transiting planet, WASP-43b, remove stellar variability, and fit a transit model to recover the planet's physical parameters with uncertainties.

a) Use lightkurve to search and download the PDCSAP flux for your target/sector (see attached notebook). Remove NaNs. Plot the full light curve. Identify the transits by eye (mark them). Report the literature period, epoch, depth, and duration you're expecting, so you have something to sanity-check against later.

b) Use wotan.flatten to remove long-term/stellar trends and to normalize your flux, i.e. to rescale your lightcurve in units of stellar flux (approximately). Mask the transits before flattening (or use wotan's transit mask / iterative clipping option), if you flatten through the transit, the detrending will partially fit out the transit itself and bias your depth. Show a plot of the light curve with and without masking to demonstrate you checked this. Justify your choice of window length / method (biweight, gp, etc.) show at least one comparison plot of two window lengths and explain why you picked the one you did. Report the out-of-transit scatter (ppm) before and after flattening.

c) Write a function transit_model(params, time) that returns model flux using batman.TransitModel that assumes an eccentricity of 0 and a quadratic limb darkening model with the following free parameters: R_p/R_{star}, the planet radius in units of stellar radii, a/R_{star}, the semi-major axis in units of stellar radii, i, the inclination, t_0, the mid-transit time, P, the orbital period, and $u_{1,}$ u_2 the limb-darkening coefficients. Plot the model at the literature parameters from the NASA Exoplanet Archives over your flattened TESS sector lightcurve.

d) Phase-folded data such that the mid-transit time is at an orbital phase of 0 and plot the model with literature values over the phase-folded data.

e) Using the fitting routine of your choice (e.g. scipy.optimize.curve_fit, emcee, dynesty, PyMC3, etc.) fit the transit model to your lightcurve. Report best-fit parameters with 1σ uncertainties from the covariance matrix or your posterior distribution. How does it compare with the values in the literature?

f) Phase-fold the data on your best-fit mid-transit time and orbital period and overplot the best-fit model. Plot residuals (data − model) and report the RMS. How good is your fit?